Dicriticals at Infinity of Keller Maps:

Theorem R and Local Obstructions

Counting Dicriticals, the Line at Infinity, First Blow-Ups, and the $m = 1$ Bridge under Its Premises

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Computation and LaTeX Source:
https://github.com/brandonmccraryresearch-cloud/2D-Jacobian-Conjecture-Workspace

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Revision note. This paper supersedes the October 1, 2026 draft *No Keller Map Admits Two Dicritical Divisors: Closure of the Two-Dicritical Line for the Planar Jacobian Conjecture*. That draft's main theorem is **false as worded**, and its claims of closure and of unconditional status are **withdrawn** here. The corrections follow the restatement note of October 13, 2026 (`branch_two_dicritical/RESTATEMENT.md`, unpublished working note): the corrected target is Theorem R, stated as **open** in §1, and every result below is labelled with its status and evidence grade. Nothing in this revision proves the planar Jacobian conjecture. The exposition is expanded, in the step-by-step style of §5 of Guccione–Guccione–Horruitiner–Valqui [1], so that each derivation shows its intermediate equations rather than only its conclusion.

Abstract

Let $F = (P, Q) \colon \mathbb{A}^2 \to \mathbb{A}^2$ be a Keller map, that is, $[P, Q] := P_x Q_y - P_y Q_x \in K^\times$ for K algebraically closed of characteristic 0, and let $\lambda P + \mu Q$ be its pencil. The October 1 draft of this paper claimed that no Keller map admits two distinct dicritical divisors, unconditionally. That claim is false: every Keller map of degree at least 2 has at least two exceptional dicriticals in total (Corollary 1.5), and the draft's E_1 degree obstruction theorem is false as stated, with an explicit Keller counterexample in §3.

The corrected target is **Theorem R** (§1), which is **open**: for every Keller map, the pencil $\lambda P + \mu Q$ has at most one dicritical divisor **at infinity**. Theorem R is equivalent to the planar Jacobian conjecture (Proposition 1.9). This paper proves, or records as sketch or fixed-degree computation, only the following steps toward it, each labelled with its grade:

- the count of dicriticals (Proposition 1.3, grade B) and the position a counterexample's dicriticals would occupy (§1.7);
- the line at infinity is not dicritical for a non-affine Keller map (Lemma 2.1, grade B), and no first blow-up at infinity is dicritical (Lemma 2.3, algebra grade A, chart step grade B), so every dicritical at infinity of a non-affine Keller map has depth at least 2;
- the $m = 1$ bridge equations collapse under two stated premises (P1) and (P2) (Theorem 4.1, equations grade A, geometric reading grade B), where (P2) is **not** derived and remains open;
- the constant-leading-form argument for $m \geq 2$ on E_k, $k \geq 2$, as a **proof sketch for free centers only** (Proposition 2.5, grade C);
- the adjacent $m = 1$ case as a computation at $d = 3$ only (grade C).

What remains open is collected in §5 (obligations O1–O6) and the tests any proof must pass (T1–T4). §6 maps each archived script to the exact statement it certifies, under the corrected statements.

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1 Introduction: setting, definitions, and the corrected statement

1.1 Context

The planar Jacobian conjecture asserts that a polynomial map $(P, Q)\colon \mathbb{C}^2 \to \mathbb{C}^2$ with $[P, Q] := P_x Q_y - P_y Q_x \in \mathbb{C}^\times$ is invertible. A *Keller map* is such a pair with constant nonzero Jacobian. This paper studies the dicritical divisors of the pencil $\lambda P + \mu Q$ of a Keller map, and in particular how many of them can lie at infinity. The October 1 draft claimed this line of work was closed. It was not, and this revision says exactly what is proved, what is only sketched or computed at fixed degree, and what is open.

Throughout, K is an algebraically closed field of characteristic 0; statements over any field of characteristic 0 reduce to its algebraic closure. We write $F = (P, Q)\colon \mathbb{A}^2 \to \mathbb{A}^2$ for a Keller map, so $[P, Q] = P_x Q_y - P_y Q_x \in K^\times$.

1.2 Standard facts used

We use the following standard facts, labelled (F1)–(F3) as in the restatement note:

(F1) F is étale. Its generic degree $e = [K(x, y) : K(P, Q)]$ is at least 1.

(F2) The *non-properness set* S_F is the set of points of the target over which F is not proper. It is either empty or a curve whose components are rational (Jelonek [3], for dominant maps $\mathbb{A}^n \to \mathbb{A}^n$).

(F3) F is an automorphism if and only if $S_F = \varnothing$. Indeed, if $S_F = \varnothing$ then F is a finite étale cover of the simply connected $\mathbb{A}^2$, hence an isomorphism; if F is not an automorphism then $e \geq 2$, since with $e = 1$ the map F is birational and étale, hence an open immersion by Zariski's main theorem, so it is injective, and then bijective (Białynicki-Birula–Rosenlicht [8]).

1.3 Compactification, the pencil, and the standard definition

Source and target are $\mathbb{A}^2 \subset \mathbb{P}^2$, with lines at infinity L_∞ and L'_∞. Let $\sigma\colon X \to \mathbb{P}^2$ be a sequence of point blow-ups over L_∞ such that $G = \bar{F} \circ \sigma\colon X \to \mathbb{P}^2$ is a morphism. The *curves at infinity* of X are the components of $X \setminus \mathbb{A}^2$: the strict transform of L_∞ and the exceptional curves.

For $c \in \mathbb{A}^2$ (target), Λ_c is the pencil whose members are $F^{-1}(\ell)$ for the lines $\ell \ni c$. Its members are the fibers of $\varphi_c = \pi_c \circ F$, where π_c is the projection from c. The paper's pencil $\lambda P + \mu Q$ is Λ_0.

Definition 1.1 (Dicritical divisor, standard definition)**.** A divisor E over the source plane is *dicritical* for Λ_c if φ_c is not constant on E. Its *degree* $m(E)$ is the degree of $\varphi_c|_E : E \to \mathbb{P}^1$. The divisor E is *at infinity* if its center on $\mathbb{P}^2$ lies on L_∞; L_∞ itself counts. Otherwise E is *affine*. The *depth* of L_∞ is 0; the exceptional curve of the blow-up at a point p has depth one more than the largest depth of the curves through p. Thus E_1 over a point of L_∞ has depth 1. A point of an exceptional curve is a *satellite* point if it lies on another component of the total transform of L_∞, and *free* otherwise.

Remark 1.2. The October 1 draft defined dicriticals as "exceptional divisors on which the pulled-back pencil has no base points." That is not the standard notion, and it is replaced by Definition 1.1 throughout this revision.

1.4 Counting dicriticals

Proposition 1.3 (The count)**.** *Status: grade B, proof by hand from* (F1)–(F3)*. Let F be a Keller map and $c \in \mathbb{A}^2 \setminus S_F$.*

(a) The affine dicriticals of Λ_c are exactly the e exceptional curves E_q of the blow-ups at the points $q \in F^{-1}(c)$. Each has degree 1.

(b) The dicriticals at infinity are the curves at infinity D of X for which $G(D)$ is a curve. Then $G(D)$ is L'_∞ or a component of $\bar{S}_F$, and

$$m(D) = \deg\big(D \to G(D)\big) \cdot \deg G(D). \tag{1}$$

(c) L'_∞ and every component of $\bar{S}_F$ are $G(D)$ for at least one such D. So the number $N_\infty(c)$ of dicriticals at infinity satisfies

$$N_\infty(c) \geq 1 + \#\{\text{components of } S_F\}. \tag{2}$$

Proof. We prove (a), (b), (c) in order, spelling out each step.

Part (a). Since $c \notin S_F$, the map F is proper over a neighbourhood of c. By (F1), F is étale, so the fiber $F^{-1}(c)$ has exactly e points, and F is a local isomorphism at each of them. Fix $q \in F^{-1}(c)$. Near q, the map φ_c is the projection from the point c composed with a local isomorphism. One blow-up at q resolves it: in the blow-up chart, the exceptional curve E_q parametrizes the directions through q, and φ_c restricts to the projection itself, hence maps E_q isomorphically onto $\mathbb{P}^1$. So E_q is dicritical of degree 1. At every other affine point, φ_c is regular, so no other affine divisor is dicritical. This gives exactly the e exceptional curves E_q, each of degree 1.

Part (b). Let D be a curve at infinity. First, $G(D)$ is not the point c: the map G sends no curve at infinity into $\mathbb{A}^2 \setminus S_F$, and $c \notin S_F$ by hypothesis. If $G(D)$ is a point different from c, then $\varphi_c = \pi_c \circ G$ is constant on D, since π_c sends every point other than c to a single point of $\mathbb{P}^1$ only after fixing the line through c — more precisely, π_c is constant on no curve through a point other than c except by collapsing it, and here $G(D)$ is a single point $\neq c$, so $\varphi_c|_D$ is that constant value. Hence D is not dicritical in this case.

If instead $G(D)$ is a curve, then either $G(D) = L'_\infty$, or $G(D)$ meets $\mathbb{A}^2$. In the second case, every point of $G(D) \cap \mathbb{A}^2$ is a limit of F along a path to infinity, so it lies in S_F; since $G(D)$ is irreducible, $G(D)$ is a component of the closure $\bar{S}_F$. Neither L'_∞ nor a component of S_F is a line through c.

Therefore the projection π_c restricted to the curve $G(D)$ has degree $\deg G(D)$, and composing with $D \to G(D)$ gives

$$m(D) = \deg(\varphi_c|_D) = \deg\big(D \to G(D)\big) \cdot \deg G(D),$$

which is (1). So D is dicritical exactly when $G(D)$ is a curve, with the stated degree.

Part (c). The map G is proper and surjective, so L'_∞ is dominated by some curve at infinity: there is D with $G(D) = L'_\infty$. By the description of S_F as $G(X \setminus \mathbb{A}^2) \cap \mathbb{A}^2$, each component of S_F is likewise dominated by a curve at infinity. Counting the distinct images L'_∞ and the components of S_F gives the lower bound (2). □

Corollary 1.4 (Automorphisms)**.** *Status: grade B. If F is an automorphism, then Λ_c has exactly two dicriticals for every c: one affine, over $F^{-1}(c)$, and one at infinity, the unique curve D with $G(D) = L'_\infty$ (here G is birational). Both have degree 1.*

Proof. If F is an automorphism, then $e = 1$ and $S_F = \varnothing$ by (F3). Part (a) of Proposition 1.3 gives exactly one affine dicritical, over the single point $F^{-1}(c)$, of degree 1. Part (c) with $S_F = \varnothing$ gives at least one dicritical at infinity, with image L'_∞; since G is birational, there is exactly one such curve, and (1) gives its degree as $1 \cdot 1 = 1$. □

1.5 The October 1 main theorem is false

Corollary 1.5 (The theorem as worded fails)**.** *Status: grade B in general; exact in the examples below. Every Keller map of degree at least 2 has at least two exceptional dicriticals in Λ_0, after a translation of the target that puts 0 in $F(\mathbb{A}^2) \setminus S_F$. An automorphism needs no translation. Hence the October 1 draft's theorem ("no two distinct dicritical divisors," with no at-infinity restriction) is false for every Keller map of degree at least 2. If only exceptional curves count as divisors of the resolution, it holds exactly for affine maps, whose dicritical at infinity is L_∞ itself; if L_∞ counts, it fails for them too.*

Proof. Let F have degree at least 2. Translate the target so that $0 \notin S_F$ and $0 \in F(\mathbb{A}^2)$; then $F^{-1}(0)$ is nonempty. By Proposition 1.3(a), each point of $F^{-1}(0)$ contributes an affine dicritical of degree 1, and by part (c) there is at least one dicritical at infinity. That is already two dicriticals in total. The example below exhibits both of them explicitly for an automorphism, where no translation is needed. □

Example 1.6 (Two dicriticals of an automorphism)**.** Take

$$P = x - y^2, \qquad Q = y. \tag{3}$$

We compute the Jacobian step by step. First the partial derivatives:

$$P_x = 1, \qquad P_y = -2y, \qquad Q_x = 0, \qquad Q_y = 1.$$

Hence

$$[P, Q] = P_x Q_y - P_y Q_x = 1 \cdot 1 - (-2y) \cdot 0 = 1 \in K^\times,$$

so $F = (P, Q)$ is a Keller map. It is an automorphism: solving $P = x - y^2$ and $Q = y$ gives $y = Q$ and $x = P + Q^2$, so the inverse is $(P, Q) \mapsto (P + Q^2, Q)$, and indeed $P + Q^2 = (x - y^2) + y^2 = x$.

The affine dicritical. The fiber over 0 is the single point $(0,0)$, since $Q = y = 0$ and $P = x - y^2 = 0$ force $x = 0$. Blow up $\mathbb{A}^2$ at the origin and use the chart

$$x = s, \qquad y = st,$$

so the exceptional divisor E is $\{s = 0\}$ with coordinate t. Substituting:

$$P = s - (st)^2 = s - s^2t^2 = s(1 - st^2), \tag{4}$$
$$Q = st. \tag{5}$$

The map φ_0 is the projection from 0, so on this chart it is the ratio

$$\varphi_0 = [P : Q] = [\, s(1 - st^2) : st \,] = [\, 1 - st^2 : t \,].$$

Restricting to the exceptional divisor, where $s = 0$, gives

$$\varphi_0|_E = [1 : t],$$

which is non-constant in t. Hence E is dicritical, of degree 1.

The dicritical at infinity. By Corollary 1.4, Λ_0 has exactly one dicritical at infinity, the unique curve D with $G(D) = L'_\infty$, also of degree 1. Together with the affine dicritical above, this is two distinct dicriticals. The October 1 theorem, which forbade two distinct dicritical divisors without any at-infinity restriction, is therefore false, already for automorphisms.

1.6 The corrected statement: Theorem R

Theorem 1.7 (Theorem R — the corrected main theorem; OPEN)**.** *Status: open, not proved in this paper or elsewhere. For every Keller map (P, Q), the pencil $\lambda P + \mu Q$ has at most one dicritical divisor at infinity.*

Proposition 1.8. *Status: grade B. Let F be a Keller map and $c \notin S_F$. Then Λ_c has at most one dicritical at infinity if and only if F is an automorphism. In that case it has exactly one, of degree 1.*

Proof. If F is an automorphism, Corollary 1.4 gives exactly one dicritical at infinity, of degree 1. Conversely, suppose Λ_c has at most one dicritical at infinity. By Proposition 1.3(c), the number of dicriticals at infinity is at least $1 + \#\{\text{components of } S_F\}$. At most one therefore forces $\#\{\text{components of } S_F\} = 0$, that is, $S_F = \varnothing$. By (F3), F is an automorphism. □

Proposition 1.9 (Theorem R is equivalent to the planar Jacobian conjecture)**.** *Status: grade B. Theorem R holds for all Keller maps if and only if the planar Jacobian conjecture holds.*

Proof. If the conjecture holds, then every Keller map is an automorphism, and Corollary 1.4 gives exactly one dicritical at infinity, in particular at most one. So Theorem R holds.

Conversely, let F be a Keller map and pick $c \notin S_F$. Such a c exists because S_F is either empty or a curve, never the whole target. Then $F - c$ is a Keller map with $0 \notin S_{F-c}$, and its pencil $\lambda(P - c_1) + \mu(Q - c_2)$ is $\Lambda_c(F)$. Applying Theorem R to $F - c$ and Proposition 1.8 to $\Lambda_c(F)$ makes F an automorphism. Since F was arbitrary, the conjecture holds. □

Remark 1.10. Proposition 1.9 sets the bar for this paper: a complete proof of the corrected theorem would be a proof of the conjecture. Everything that follows is therefore measured against that bar, and no step below is presented as closing it.

1.7 Where a counterexample's dicriticals would sit

Let F be a Keller map that is not an automorphism, and let $c \notin S_F$. Then Λ_c has:

- $e \geq 2$ affine dicriticals, by (F3) and Proposition 1.3(a);
- at least one E_∞ with $G(E_\infty) = L'_\infty$, of degree $\deg(E_\infty \to L'_\infty)$;
- at least one E_S over each component C of S_F, of degree $\deg(E_S \to C) \cdot \deg C$, by (1).

E_∞ always exists, so Theorem R amounts to showing that E_S cannot exist. The two halves of the October 1 draft, read against this picture, were: the first half claimed there is no dicritical of degree $m \geq 2$ for any Keller map, which applied to E_S would already exclude every counterexample whose S_F has a component of degree at least 2; the second half claimed there are no two dicriticals of degree 1, which would handle the rest. Either half, to succeed, must apply to E_∞ and E_S wherever they sit in the resolution tree: over any point of L_∞, at any depth, and after free or satellite blow-ups. Sections 2–4 record how far each half actually goes.

2 Dicriticals at infinity of small depth

This section proves the two depth 0 and depth 1 steps toward Theorem R, each with every intermediate computation shown, and then records the constant-leading-form argument in its honest state: a sketch for free centers only. The conclusion of this section is that every dicritical at infinity of a non-affine Keller map has depth at least 2. Nothing here touches depth 2 or beyond, except for the sketch in §2.3, whose limits are stated there.

2.1 The line at infinity

Lemma 2.1 (The line at infinity)**.** *Status: grade B, proof by hand; the binary-form step is executed in* ***`base_point_at_infinity_check.py`****, check* 1*. For a non-affine Keller map, L_∞ is not dicritical for any Λ_c.*

Proof. Let $d = \max(\deg P, \deg Q) \geq 2$, the non-affine case. Write P_d and Q_d for the homogeneous components of degree d of P and Q (zero if the corresponding degree is smaller than d). The restriction $\varphi_c|_{L_\infty}$ is the rational function P_d/Q_d on $L_\infty \cong \mathbb{P}^1$, where the point c contributes only a constant shift that does not affect constancy. We show this ratio is constant, in two cases.

Case 1: unequal degrees. Suppose $\deg P \neq \deg Q$, say $\deg P > \deg Q$ (the other case is symmetric). Then on L_∞, the term of highest degree in the pencil is P_d alone, so P_d/Q_d is ∞ wherever Q_d vanishes to lower order, and the restriction is the constant ∞. Explicitly: homogenizing, the leading part of a member $\lambda P + \mu Q$ at infinity is λP_d (since $\deg Q < d$), so the pencil has the single point at infinity determined by P_d as its value, independent of the member. The restriction is constant, so L_∞ is not dicritical. The case $\deg Q > \deg P$ gives the constant 0 by the same computation.

Case 2: equal degrees. Suppose $\deg P = \deg Q = d$. The Jacobian $[P, Q]$ is a nonzero constant, so in particular its homogeneous component of degree $2d - 2$ vanishes. That component is exactly $[P_d, Q_d]$, because differentiating lowers degree by 1 in each factor and the bracket of the top components is the top component of the bracket:

$$[P, Q]_{2d-2} = [P_d, Q_d] = 0. \tag{6}$$

So it suffices to prove the following binary-form lemma.

Lemma 2.2 (Proportionality of binary forms with vanishing bracket)**.** *Let f, g be binary forms (homogeneous polynomials in two variables) of the same degree ≥ 1. If $[f, g] = 0$, then $f = rg$ for some constant $r \in K$.*

Proof of Lemma 2.2, step by step. Work in the rational function field and set $h = f/g$, a rational function of degree 0, meaning $h(tx, ty) = h(x, y)$ for all t. Differentiating this homogeneity relation with respect to t at $t = 1$ gives Euler's relation for h:

$$x h_x + y h_y = 0. \tag{7}$$

Next, the bracket is bilinear and satisfies the quotient rule in the first argument:

$$[h, g] = \frac{[f, g]}{g} - \frac{f[g, g]}{g^2} = \frac{[f, g]}{g} = 0,$$

using $[g, g] = 0$ and the hypothesis $[f, g] = 0$. Writing the bracket out,

$$h_x g_y - h_y g_x = 0. \tag{8}$$

Now suppose $(h_x, h_y) \neq (0, 0)$. Equation (8) says the vector (h_x, h_y) is proportional to (g_x, g_y): both are orthogonal to the same direction, so there is a scalar λ with $(g_x, g_y) = \lambda(h_x, h_y)$ wherever the vectors are defined. Equation (7) says (h_x, h_y) is proportional to $(-y, x)$: indeed, $x h_x + y h_y = 0$ means (h_x, h_y) is orthogonal to (x, y), and in two variables the orthogonal direction is spanned by $(-y, x)$. Combining the two proportionalities, (g_x, g_y) is proportional to $(-y, x)$, so

$$x g_x + y g_y = 0.$$

But Euler for the homogeneous form g of degree $\deg g \geq 1$ says

$$x g_x + y g_y = (\deg g) \cdot g \neq 0.$$

This is a contradiction. Hence the supposition is false, and $(h_x, h_y) = (0, 0)$. In characteristic 0, a rational function with vanishing partial derivatives is constant, so $h = r$ for some $r \in K$, that is, $f = rg$. □

Returning to Case 2: Lemma 2.2 applied to $f = P_d$, $g = Q_d$ gives $P_d = rQ_d$, so $\varphi_c|_{L_\infty} = P_d/Q_d = r$ is constant. Hence L_∞ is not dicritical. This completes both cases. □

2.2 First blow-ups at infinity

Lemma 2.3 (First blow-ups at infinity)**.** *Status: algebra grade A, in* ***`lean/TwoDicritical/E1Keller.lean`*** *(****`e1_lead_proportional`****,* ***`jac2_eq_zero_of_natDegree_zero`****); chart step grade B, executed in* ***`lean/check_e1_index.py`****, check 5, not Lean-formalized. For a Keller map, the exceptional curve of the blow-up at any point of L_∞ is not dicritical. This holds for every degree m, not only for $m = 2$.*

Proof. A linear change of coordinates keeps the Keller condition (the Jacobian transforms by the constant determinant of the change), so we may move the point of L_∞ to $[1 : 0 : 0]$. Let E_1 be the exceptional curve of the blow-up at this point, with the chart coordinate t_1 on

E_1 as in `lean/check_e1_index.py`. The proof has three steps: the restriction formula, the Wronskian computation, and the degenerate cases.

Step 1: the restriction formula. Suppose P and Q have the same x-degree $n \geq 1$, with x^n-coefficients $A, D \in K[y]$, so that

$$P = A(y)\,x^n + (\text{terms of } x\text{-degree} < n), \qquad Q = D(y)\,x^n + (\text{terms of } x\text{-degree} < n). \tag{9}$$

In the E_1 chart, the ratio P/Q restricts to E_1 as

$$\left.\frac{P}{Q}\right|_{E_1} = \frac{A(1/t_1)}{D(1/t_1)}. \tag{10}$$

The reason is that on E_1 only the top x-degree terms survive: lower x-degree terms vanish to higher order along E_1, and the chart coordinate t_1 inverts the y-variable, which is why A and D are evaluated at $1/t_1$. If instead the x-degrees of P and Q differ, say $\deg_x P > \deg_x Q$, then the numerator dominates and the restriction is the constant ∞; if $\deg_x P < \deg_x Q$, it is the constant 0. In both differing cases E_1 is already not dicritical, so only the equal-degree case remains.

Step 2: the Wronskian computation. Differentiate the expansions (9) term by term. The top x-degree contributions are:

$$P_x = nA(y)\,x^{n-1} + (x\text{-degree} < n-1), \tag{11}$$
$$P_y = A'(y)\,x^n + (x\text{-degree} < n), \tag{12}$$
$$Q_x = nD(y)\,x^{n-1} + (x\text{-degree} < n-1), \tag{13}$$
$$Q_y = D'(y)\,x^n + (x\text{-degree} < n). \tag{14}$$

Now form the Jacobian and collect the top x-degree. The product P_xQ_y has top x-degree $(n-1)+n = 2n-1$, with coefficient

$$(nA)\cdot D' = nAD'.$$

The product P_yQ_x likewise has top x-degree $n+(n-1) = 2n-1$, with coefficient

$$A'\cdot(nD) = nA'D.$$

Every other product of a term from P_x, P_y with a term from Q_x, Q_y has x-degree strictly less than $2n-1$: lowering the x-degree of either factor by at least 1 lowers the total by at least 1. Therefore the x^{2n-1}-coefficient of $[P,Q]$ is exactly

$$[P,Q]_{x^{2n-1}} = n\left(AD' - A'D\right). \tag{15}$$

Since (P,Q) is Keller, $[P,Q]$ is a nonzero constant, so every coefficient of a positive power of x in $[P,Q]$ vanishes. In particular the coefficient (15) vanishes, and since $n \geq 1$,

$$AD' - A'D = 0.$$

This is the vanishing of the Wronskian of A and D. In characteristic 0 it forces A and D to be proportional: indeed, where $D \neq 0$,

$$\left(\frac{A}{D}\right)' = \frac{A'D - AD'}{D^2} = 0,$$

so A/D is constant, say $A = rD$ for $r \in K$. (This is the Lean lemma `e1_lead_proportional`.) Substituting into the restriction formula (10) gives

$$\left.\frac{P}{Q}\right|_{E_1} = \frac{rD(1/t_1)}{D(1/t_1)} = r,$$

a constant. Hence E_1 is not dicritical in the equal-degree case either.

Step 3: the degenerate case. If both x-degrees are 0, then $P, Q \in K[y]$ are functions of y alone, so $P_x = Q_x = 0$ and $[P, Q] = 0$. This contradicts the Keller condition $[P, Q] \in K^\times$ (Lean lemma `jac2_eq_zero_of_natDegree_zero`). So this case cannot occur for a Keller map.

Since a linear change of coordinates moves any point of L_∞ to $[1 : 0 : 0]$ while preserving the Keller condition, the same conclusion holds over every point at infinity. □

Corollary 2.4 (Depth at least 2)**.** *Status: follows from Lemmas 2.1 and 2.3. Every dicritical at infinity of a non-affine Keller map has depth at least* 2.

Proof. Depth 0 is L_∞ itself, excluded by Lemma 2.1 for non-affine maps. Depth 1 curves are the exceptional curves of blow-ups at points of L_∞, excluded by Lemma 2.3. The smallest remaining depth is 2. □

2.3 The constant-leading-form argument: a sketch for free centers only

Proposition 2.5 (Constant leading form — proof sketch)**.** *Status: grade C, proof sketch; free centers only. This is not a proof for satellite chains or for depth* ≥ 2 *in general (obligation O3). Suppose* E_k *(*$k \geq 2$*) is dicritical of order* $m \geq 1$ *for the pencil. Then after the standard normalization, the leading coefficient satisfies* $\deg_{u'} U(u', 0) = 0$.

Proof sketch. Let E_k ($k \geq 2$) be an exceptional divisor in the resolution, with local coordinates (u', v') in which $E_k = \{v' = 0\}$. For a polynomial $S \in K[x, y]$ of degree d, write the pullback as

$$S = v'^{-v}\, U(u', v') \cdot (\text{monomial in } u'), \tag{16}$$

where $v = v_{E_k}(S)$ is the order of vanishing and U is a unit (nonzero at the center). The dicritical condition of order m requires the u'-degree of the pencil restriction to equal m.

Track the unit U through the blow-up sequence, one blow-up at a time. Consider a single blow-up at a center with $u_0 \neq 0$ (a *free* center: the center does not lie on another component of the total transform, so the local coordinate u' is nonzero there). In the blow-up chart, substituting the center shifts the leading coefficient by the constant factor u_0^{-2}: schematically, if before the blow-up the leading term is $c \cdot u'^j$, then after the blow-up it is $c\, u_0^{-2} \cdot u''^j$ in the new coordinate u'', with the same exponent j and a nonzero constant multiple. The point is that multiplication by the nonzero constant u_0^{-2} changes no u'-degree: the exponent j is carried through unchanged, and no new u'-dependence is created, because the center coordinate u_0 is a nonzero constant rather than a function of u'.

Repeating this step along the chain: each blow-up at a center with $u_0 \neq 0$ multiplies the leading coefficient by a nonzero constant and preserves its u'-degree. A direct computation in the normalized chart then shows the u'-degree of $U(u', 0)$ is 0. The verification of the degree count is in `general_constant_form.py` (§6).

Where the sketch stops. The step above uses $u_0 \neq 0$ at every blow-up. If a center lies on another component of the total transform (a satellite center), then the corresponding coordinate vanishes at the center, the factor is no longer a nonzero constant, and the degree-tracking argument as written does not go through. The sketch therefore covers free chains only. It does not prove that no E_k ($k \geq 2$) is dicritical of order $m \geq 2$ in general, and it is not presented as doing so: the general $m \geq 2$ statement at depth ≥ 2 is obligation O3 in §5. The middle-divisor (E_2) and leaf cases (E_3, E_4) are checked separately as computations in `middle_e2.py`, `leaf_e3.py`, and `leaf_e4.py`; those are fixed instances, not the general argument. □

Remark 2.6. In the October 1 draft, Proposition 2.5 was followed by a corollary claiming no E_k ($k \geq 2$) is dicritical of order $m \geq 2$ for any Keller map, and that corollary was then used to close the $m \geq 2$ half of the main theorem. That inference is withdrawn: a sketch for free centers does not yield the general corollary, and the general corollary is not claimed anywhere in this revision.

3 The first exceptional divisor: a false obstruction and its correction

This section does two things the October 1 draft got backwards. First, it exhibits the draft's E_1 degree obstruction as false, shows the exact step where its proof goes wrong, and computes the counterexample in full. Second, it records the corrected statement, which was already proved as Lemma 2.3 in §2.2: first blow-ups at infinity are never dicritical for a Keller map, for every degree m. The correction is strictly stronger than the draft's claim in scope (every m, not only $m = 2$) and strictly weaker in what it asserts about E_1 itself (it does not claim anything about a hypothetical $m = 2$ open subset, which, as the counterexample shows, contains Keller pairs).

3.1 What the October 1 draft claimed

The draft's E_1 degree obstruction theorem read, in substance: let P, Q lie in the E_1 $m = 2$ open subset, meaning $v_{E_1} = -2$ and both pencil restrictions have exact degree 2; then $[P, Q]$ is never a nonzero constant. Its proof argued as follows: every monomial $x^J y^K$ allowed by the order condition $v_{E_1} \geq -2$ satisfies $J + K \geq 2$; hence P_x, P_y, Q_x, Q_y have no constant term; hence $[P, Q]$ has minimum total degree 2 and cannot be a nonzero constant. Every step after the first sentence is correct arithmetic. The first sentence is false, and this subsection shows why, step by step.

3.2 The order condition, computed correctly

Lemma 3.1 (The E_1 order condition)**.** *In the E_1 chart, the valuation satisfies*

$$v_{E_1}(x^i y^j) = -i. \tag{17}$$

Consequently $v_{E_1} \geq -2$ holds for a monomial $x^i y^j$ if and only if $i \leq 2$, that is, if and only if the x-degree is at most 2. It says nothing about the total degree $i + j$.

Proof. Equation (17) is the normalization of the E_1 valuation used in the E_1 chart (see `lean/check_e1_index.py` and `lean/TwoDicritical/E1Keller.lean`): the variable x has a

pole of order 1 along E_1, contributing -1 per power of x, while y contributes 0. Hence for the monomial $x^i y^j$,

$$v_{E_1}(x^i y^j) = i \cdot v_{E_1}(x) + j \cdot v_{E_1}(y) = i \cdot (-1) + j \cdot 0 = -i.$$

The inequality $v_{E_1} \geq -2$ is therefore

$$-i \geq -2 \quad \Longleftrightarrow \quad i \leq 2.$$

In words: the condition bounds the x-degree by 2. It does not bound j, and it does not bound the total degree $i + j$ from below: the monomial y itself, with $i = 0$ and $j = 1$, satisfies $v_{E_1}(y) = 0 \geq -2$ while having total degree 1. The draft's claim that every allowed monomial has $J + K \geq 2$ is thus false on its face, once the valuation is written out. □

3.3 How the mix-up happened

The draft's error is an index mix-up, identified in the Lean analysis (`lean/README.md`, item 2). The computation `e1_solitary.py` indexes its unknowns $p[J, K]$ by the homogeneous monomial $X^{d-J-K} Y^J Z^K$, that is, by the affine monomial

$$x^{d-J-K} y^J.$$

The draft's proof, and the scripts `e1_keller_search.py` and the Lean file `lean/TwoDicritical/E1Obstruction.lean`, read $p[J, K]$ as the monomial $x^J y^K$ instead. Under that misreading, the condition on the indices becomes a condition on the total degree $J+K$, and the draft concluded that every allowed monomial has total degree at least 2. At $d = 4$, both index sets have 12 monomials, so the count agreed and the mix-up was hidden: the 12-dimensional span at $d = 4$ is real, but it is a span of monomials of x-degree at most 2, not of total degree at least 2. The file `lean/TwoDicritical/E1Obstruction.lean` proves a true lemma about the misread set of monomials (if every monomial of P and Q has total degree at least 2, then $[P, Q]$ has no constant term), but that set is not the E_1 condition, so the lemma does not prove anything about E_1.

3.4 The counterexample, computed in full

Example 3.2 (A Keller pair in the draft's E_1 open subset)**.** *Status: grade A in Lean (**`lean/TwoDicritical/E1Keller.lean`**, **`thm_e1_false`**); the computation below is exact.* Set

$$P = (x + y^2)^2 + y, \qquad Q = P + x + y^2. \tag{18}$$

We expand P first, term by term:

$$P = x^2 + 2xy^2 + y^4 + y.$$

Its partial derivatives are:

$$P_x = 2x + 2y^2 = 2\,(x + y^2), \tag{19}$$

$$P_y = 4xy + 4y^3 + 1 = 4y\,(x + y^2) + 1. \tag{20}$$

Now compute $[P, Q]$. Since $Q = P + x + y^2$ and $[P, P] = 0$, bilinearity of the bracket gives

$$[P, Q] = [P, x] + [P, y^2].$$

Each bracket is computed separately. For $[P, x]$: x has derivatives 1 and 0, so

$$[P, x] = P_x \cdot 0 - P_y \cdot 1 = -P_y = -\big(4y(x + y^2) + 1\big).$$

For $[P, y^2]$: y^2 has derivatives 0 and $2y$, so

$$[P, y^2] = P_x \cdot 2y - P_y \cdot 0 = 2yP_x = 2y \cdot 2(x + y^2) = 4y(x + y^2).$$

Adding the two contributions, the $4y(x + y^2)$ terms cancel exactly:

$$[P, Q] = -\big(4y(x + y^2) + 1\big) + 4y(x + y^2) = -1 \in K^\times. \tag{21}$$

So (P, Q) is a Keller pair.

It lies in the draft's E_1 $m = 2$ open subset at $d = 4$: its x-degree is 2 (the term x^2), so $v_{E_1} = -2$ by Lemma 3.1, and each pencil restriction is the function $C(t_1) = t_1^2$ of exact degree 2 (this is the membership statement formalized as `thm_e1_false` in `lean/TwoDicritical/E1Keller.lean`, with the open subset denoted `E1Open4` there). Yet $[P, Q] = -1$ is a nonzero constant, contradicting the draft's theorem directly.

Two further observations pin down the failure. First, P contains the monomial y, of total degree 1: the draft's claimed lower bound $J + K \geq 2$ fails for a monomial that is actually present. Second, P/Q restricts to 1 on E_1 for this pair, so E_1 is not dicritical for them at all: the draft's "open subset" does not even capture an $m = 2$ dicritical. The draft's proof argued about the wrong monomials, and its conclusion is false.

Remark 3.3 (Withdrawn theorem). The draft's E_1 degree obstruction theorem and its corollary ("no dicritical of order $m \geq 2$ exists for any Keller map") are both withdrawn. The corollary depended on the false theorem for the E_1 case and on the free-only sketch of Proposition 2.5 for the E_k, $k \geq 2$, case; neither leg supports it as stated.

3.5 The corrected E_1 statement

The intended conclusion of the draft's E_1 argument — that the first blow-up cannot be dicritical for a Keller map — is true, and it is Lemma 2.3 of §2.2, proved there for every degree m. To recap the logic in one place: the restriction of P/Q to E_1 is $A(1/t_1)/D(1/t_1)$ when the x-degrees agree (and 0 or ∞ when they differ), and the Keller condition forces the Wronskian $AD' - A'D$ to vanish via the x^{2n-1}-coefficient (15), hence $A = rD$ and the restriction is the constant r. The algebra is Lean-grade-A; the chart step is executed, not Lean-formalized (obligation O5). No $m = 2$ open subset, and no total-degree argument, is used anywhere in the corrected proof.

4 The $m = 1$ bridge, under its premises

This section keeps the October 1 draft's Vandermonde and block-rank argument, which is correct as linear algebra, but restates its scope exactly. The argument proves a collapse of a system of equations. Whether that system is the system of two $m = 1$ dicriticals depends on two premises, (P1) and (P2), stated below. Premise (P2) is not derived anywhere in this paper; it is an open obligation (O2). The theorem of this section is therefore a theorem about equations under premises, not an unconditional geometric statement, and its corollaries are fenced accordingly.

4.1 The chart and the two premises

Read through its chart, the bridge concerns the following geometry. The paper's pullback is

$$S^h\big(1, b, b^2(a + \tilde{a}b)\big),$$

the chart at the point $w_2 = a$ of E_2 over $[1 : 0 : 0]$, where $w_2 = x/y^2$. The two centers $a_3 \neq a_4$ are points of this E_2. The bridge argument rests on:

(P1) the two dicriticals lie over two distinct *free* points of this E_2;
(P2) every member of the pencil has pole order at most 1 on them.

Premise (P1) is part of the geometric reading of the equations: the chart above is the chart at a free point of E_2, and the argument is set in that chart. Premise (P2) is a hypothesis on the members, not a consequence of the definitions used here: the paper defines m as the degree of $\varphi_c|_E$, and degree 1 does not force pole order 1 (`lean/check_satellite.py`, check 7). Deriving (P2) from the definition of m, or supplying a separate argument for higher pole order, is obligation O2 in §5. Nothing in this section assumes (P2) silently: every statement that needs it says so.

4.2 Setup of the equations

Consider two distinct centers $a_3 \neq a_4$ (both nonzero), each requiring order $\mathrm{ord}_b(S) \geq 2d - 1$ (the $m = 1$ condition, under (P2)). For $S = \sum p_{JK} x^J y^K$, the pullback at center a is

$$S_a(b) = \sum_{J,K} p_{JK}\,(a + \tilde{a}b)^{d-J-K}\, b^{2d-2J-K}, \tag{22}$$

and we require the coefficients of $b^0, \ldots, b^{2d-2}$ to vanish at $a = a_3$ and at $a = a_4$. The exponent $d - J - K$ is the remaining degree after the monomial $x^J y^K$ is factored out, and the exponent $2d - 2J - K$ is the b-order contributed by that monomial in the chart; the condition $\mathrm{ord}_b(S) \geq 2d - 1$ is exactly the vanishing of all coefficients below b^{2d-1}.

Theorem 4.1 (Bridge collapse, under (P1) and (P2))**.** *Status: equations grade A, in* ***`lean/TwoDicritical/BridgeCollapse.lean`****; geometric reading grade B, executed in* ***`lean/check_satellite.py`****. For arbitrary distinct nonzero a_3, a_4 and arbitrary $d \geq 2$, the simultaneous conditions* $\mathrm{ord}_b(S) \geq 2d - 1$ *at both centers force $p_{JK} = 0$ whenever $J \geq 1$. The nullspace is exactly* $\mathrm{span}\{1, y\}$.

In particular, every P, Q in the span is a function of y alone, so $[P, Q] \equiv 0$: no Keller pair exists in the system of equations. Whether this rules out two geometric $m = 1$ dicriticals depends on (P1) and (P2); see §4.5.

4.3 Descending induction on weighted degree

Order monomials by the weighted degree $W = 2J + K$, descending. For fixed W, write the variables as (J_k, K_k), $k = 1, \ldots, n$, ordered by descending J_k. Define

$$e_k = d - J_k - K_k = d - W + J_k. \tag{23}$$

Since W is fixed and the J_k are strictly decreasing by the ordering, the e_k are strictly decreasing, hence distinct. This distinctness is the whole input to the rank lemma below, so we record it explicitly: for $k < l$,

$$e_k - e_l = J_k - J_l > 0.$$

The equations for weight W come from the pairs (t, i) with $2d - t + i = W$, where t is the b-power and i the $\tilde{a}$-power, at both centers. To see why, expand the factor $(a + \tilde{a}b)^{e_k}$ in (22) by the binomial theorem, term by term:

$$(a + \tilde{a}b)^{e_k} = \sum_{i \geq 0} \binom{e_k}{i} a^{e_k - i} (\tilde{a}b)^i.$$

Multiplying by $b^{2d - 2J_k - K_k}$ and using $W = 2J_k + K_k$, the monomial $p_{J_k K_k}$ contributes to the coefficient of b^t exactly the terms with $2d - W + i = t$, that is, $i = t - (2d - W)$. Collecting over k gives the (t, i, a) equation:

$$\sum_{k=1}^{n} p_{J_k K_k} \binom{e_k}{i} a^{e_k - i} = 0, \qquad a \in \{a_3, a_4\}. \tag{24}$$

Crucially, the equations (24) for weight W involve only variables of weight exactly W at the top: variables of weight $> W$ contribute to higher b-powers or have already been eliminated, as the induction below makes precise.

Lemma 4.2 (Block full rank). *For distinct $e_1, \dots, e_n$ and $m \geq n$, the $2m \times n$ matrix with entries $\binom{e_k}{i} a^{e_k - i}$ ($i = 0, \dots, m - 1$, $a \in \{a_3, a_4\}$) has rank n.*

Proof. We reduce the matrix in three explicit steps, each preserving rank.

Step 1: factor the columns and rows. Factor column k by $a^{e_k} \neq 0$ (note a is a_3 or a_4, both nonzero by hypothesis) and row (i, a) by $a^{-i} \neq 0$. Rank is preserved under nonzero column and row scalings. After this step, the entry in row (i, a), column k is $\binom{e_k}{i}$, independent of a. The two centers now give two copies of the same rows, so it suffices to show that the $m \times n$ matrix $[\binom{e_k}{i}]$ ($i = 0, \dots, m - 1$) has rank n; taking the first n rows ($i = 0, \dots, n - 1$) suffices, since $m \geq n$.

Step 2: triangularize by row operations. For fixed i, the binomial coefficient $\binom{e}{i}$, viewed as a function of e, is a polynomial in e of exact degree i:

$$\binom{e}{i} = \frac{e(e-1)\cdots(e-i+1)}{i!} = \frac{1}{i!} e^i + (\text{terms of degree} < i \text{ in } e).$$

Hence row i of the matrix $[\binom{e_k}{i}]$ equals $\frac{1}{i!}$ times the row $[e_k^i]$ plus a linear combination of the rows $[e_k^j]$ for $j < i$. Proceeding from $i = 0$ upward, subtract suitable multiples of earlier rows from row i to kill all lower-degree terms: row 0 is already $[1]$; row 1 is $[e_k]$ + (constant row), so subtracting the constant multiple of row 0 leaves $[e_k]$; row 2 is $\frac{1}{2}[e_k^2]$ + (combination of rows 0, 1), so subtracting leaves $\frac{1}{2}[e_k^2]$; and so on. These are elementary row operations, so they preserve rank and multiply the determinant by 1. After all n steps, the matrix is the scaled Vandermonde matrix $[e_k^i / i!]$ ($i = 0, \dots, n - 1$).

Step 3: the Vandermonde determinant. The determinant of $[e_k^i]$ is the classical Vandermonde product $\prod_{j<k}(e_k - e_j)$, nonzero because the e_k are distinct (shown above from the ordering by J_k). Factoring $1/i!$ out of row i gives

$$\det\left[\binom{e_k}{i}\right] = \frac{\prod_{j<k}(e_k - e_j)}{\prod_{i=0}^{n-1} i!} \neq 0. \tag{25}$$

A nonzero $n \times n$ minor means the matrix has rank n. The $n = 3$ case is verified symbolically in `m1_block_rank.py`, giving $-(e_1 - e_2)(e_1 - e_3)(e_2 - e_3)/2$, which agrees with (25) since $\prod_{i=0}^{2} i! = 0! \cdot 1! \cdot 2! = 2$ and the sign is the ordering of the factors. This completes the proof. □

Proof of Theorem 4.1. Descend on W from $2d$ to 2, maintaining the inductive hypothesis that all variables with weight $> W$ are zero. The base case is $W = 2d$: there are no variables of weight $> 2d$ (since $W = 2J + K \leq 2d$ for monomials of degree at most d), so the hypothesis holds vacuously.

For the inductive step at weight W, the equations (24) for weight W involve only variables of weight $\geq W$: a variable of weight $W' > W$ contributes to b-powers above those in the weight-W system, and by the inductive hypothesis all such variables are already 0. Hence the weight-W system is exactly the block system of Lemma 4.2, which has full column rank n. Therefore $p_{J_k K_k} = 0$ for all k, that is, all variables of weight W vanish. This restores the hypothesis for the next lower weight.

Descending to $W = 2$ eliminates every variable with $J \geq 1$: any monomial with $J \geq 1$ has $W = 2J + K \geq 2$, and the descent covers all weights $W \geq 2$. The only monomials with $W < 2$ are $(J, K) = (0, 0)$ with $W = 0$ and $(J, K) = (0, 1)$ with $W = 1$. These are never constrained by the descent, so the nullspace is exactly $\mathrm{span}\{1, y\}$. □

4.4 Computational verification

The induction is verified computationally, at the fixed degrees and instances stated (these are checks of the linear algebra, not replacements for the proof above):

- Symbolic nullspace at general $a_3 \neq a_4$: $d = 2$ gives nullity 2; $d = 3$ gives nullity 2 (`m1_bridge_general.py`).
- Numeric nullspace at $a_3 = 2$, $a_4 = 5$: nullity 2 for $d = 2, \ldots, 8$ (`m1_induction.py`).
- Block-triangular structure exhibited at $d = 6$ (`m1_triangular.py`).
- Vandermonde determinant verified symbolically (`m1_block_rank.py`).

4.5 Scope of the bridge result

Under premises (P1) and (P2), the bridge has the following consequences, and no others are claimed:

- With a single center imposed, the same block argument gives the larger nullspace $\mathrm{span}\{1, y, x - ay^2\}$ (`lean/TwoDicritical/OneCenter.lean`, `bridge_one_center`). So one such dicritical already forces degree at most 2: no pencil of degree at least 3 has even one dicritical of this chart type.
- In degree 2, the automorphism $(x - ay^2, y)$ has exactly one such dicritical, consistent with Theorem R (it has one dicritical at infinity, as Corollary 1.4 requires, and no second).

- The bridge as proved here says nothing about pencils of degree at least 3 beyond the one-center remark above, because (P2) is not derived: a dicritical of degree 1 with pole order greater than 1 is not governed by the system (24). Nor does it cover dicriticals over satellite points, over different points of L_∞, or at different depths; those are obligation O1.

4.6 The adjacent case: a computation at $d = 3$ only

The adjacent $m = 1$ case — two dicriticals sharing a curve in the resolution tree rather than lying over two free points of one E_2 — is treated in this paper only by computation at $d = 3$ (`e3_adjacent.py`, `routeB_cancellation.py`), covering the E_2–E_3 adjacent pair at that degree. *Status: grade C, fixed-degree computation.* There is no argument here for other degrees; the general adjacent case is obligation O4 in §5. In the October 1 draft, this computation was cited as closing the adjacent case for every degree. That citation is withdrawn.

5 What remains open

This section replaces the October 1 draft's "Proof of the main theorem," which assembled the false E_1 theorem, the free-only constant-form sketch, the premises-dependent bridge, and the $d = 3$ adjacent computation into a contradiction, via a case split ("an adjacent pair or a bridge") that was asserted, not proved. No such assembly is attempted here. The obligations below are what a proof of Theorem R still needs; the tests after them are checks any proposed proof must pass.

5.1 Obligations

O1. The case split. The positions of two dicriticals in a resolution tree are more varied than "an adjacent pair or a bridge": dicriticals can lie over *different points* of L_∞ (a pencil can have several base points at infinity in general, as the non-Keller map (x, xy) shows, and whether a Keller counterexample's pencil can must be settled); over *satellite points* (the point $a = \infty$ of E_2 in $(x + y^3, y)$, and deeper ones); and at *different depths*. A proof must cover every configuration of E_∞ and E_S from §1.7.

O2. Premise (P2). Derive pole order at most 1 from the definition of m, or restate the bridge with (P2) as a hypothesis and supply a separate argument for higher pole order. Until then, the bridge governs only the system (22)–(24) under (P1) and (P2).

O3. $m \geq 2$ at depth ≥ 2. Prove the constant-leading-form statement of Proposition 2.5 for every chain of centers, satellite ones included. Today there is a sketch for free chains only.

O4. The adjacent case for every degree, not only $d = 3$.

O5. Two small formalizations: the chart step of Lemma 2.3 and the binary-form step of Lemma 2.1 in Lean. The algebra of Lemma 2.3 is formalized; the chart step is executed, not formalized.

O6. The Keller condition must be used globally. Theorem R is false without it. The map (x, xy), with Jacobian x, has two dicriticals at infinity in $\Lambda_{(1,1)}$, over $[1 : 0 : 0]$ and $[0 : 1 : 0]$; the second maps onto the line $u = 0$ of non-proper values (`restatement_checks.py`, check 2). Any argument that never uses $[P, Q] \in K^\times$ cannot prove Theorem R. This is a standing check on every step above: Lemmas 2.1 and 2.3 use the Keller condition (vanishing of $[P_d, Q_d]$, vanishing of the Wronskian coefficient), while the

bridge's linear algebra and the constant-form sketch must be read with their premises attached.

5.2 Tests any proof must pass

T1. Automorphisms have exactly one dicritical at infinity, of degree 1 (Corollary 1.4), but its position varies: over the free point a of E_2 for $(x - ay^2, y)$ (where $P/Q = \tilde{a}/a$ there), and over the satellite point $a = \infty$ of E_2, at depth 5, for $(x + y^3, y)$. A lemma that rules out a dicritical in either position is false.

T2. The identity. Its dicritical at infinity is L_∞ itself; Lemma 2.1 excludes only non-affine maps.

T3. Non-Keller maps can have two dicriticals at infinity (O6).

T4. Every Keller map of degree at least 2 has two exceptional dicriticals in total (Corollary 1.5), so only the count *at infinity* can be bounded. Any argument that bounds the total count is refuted by Example 1.6.

6 Script-to-arithmetic correspondence

Each script below was actually executed; outputs are archived in `outputs/`. The table maps every script to the exact paper statement it certifies, *under the corrected statements of this revision*. The grade column follows the restatement note, §8: A means machine-checked, B means a proof by hand from stated facts or an exact computation outside Lean, C means a sketch or a computation at fixed degrees. A script that certified a withdrawn statement is listed with what it actually certifies now, not with the draft's reading of it.

Script	Paper locus	Certifies	Grade
`general_constant_form.py`	Prop. 2.5	Leading-unit degree count, free chains only	C
`middle_e2.py`	Prop. 2.5	E_2 instance of the sketch	C
`leaf_e3.py`	Prop. 2.5	E_3 leaf instance of the sketch	C
`leaf_e4.py`	Prop. 2.5	E_4 leaf instance of the sketch	C
`e1_solitary.py`	§3	12-monomial count at $d = 4$ (indexing as computed)	C
`e1_global.py`	§3	Explicit pair, $v = -2$, deg = 2 (membership data)	C
`e1_keller_search.py`	§3	Searched the misread monomial set; superseded	—
`m1_bridge_general.py`	Thm. 4.1	Symbolic nullspace, $d = 2, 3$	B
`m1_induction.py`	§4	Nullity 2 for $d = 2, \ldots, 8$	C
`m1_triangular.py`	§4	Block-triangular structure at $d = 6$	C
`m1_block_rank.py`	Lem. 4.2	Vandermonde determinant $\neq 0$	B
`at_degree_bound.py`	Prop. 2.5	Finite bridge table, $m \leq 4$	C
`at_degree_bound_m5.py`	Prop. 2.5	Finite bridge table, $m \leq 5$	C
`mixed_poles.py`	Prop. 2.5	Mixed $(2, 3)$ bridge instance	C
`depth5_global.py`	Prop. 2.5	Depth-5 enumeration	C
`depth6_enum.py`	Prop. 2.5	Depth-6 enumeration	C
`routeB_cancellation.py`	§4	Adjacent $m = 1$ at $d = 3$ only	C
`e3_adjacent.py`	§4	E_2–E_3 adjacent pair at $d = 3$	C

The Lean evidence, which is not a script in the table above, is:

- `lean/TwoDicritical/E1Keller.lean`: `thm_e1_false` (the draft's E_1 theorem is false, Example 3.2); `e1_lead_proportional` and `jac2_eq_zero_of_natDegree_zero` (the algebra of Lemma 2.3); grade A for the algebra.

- `lean/TwoDicritical/BridgeCollapse.lean` and `lean/TwoDicritical/OneCenter.lean`: the bridge equations and the one-center span (Theorem 4.1, §4.5); grade A for the equations.
- `lean/check_e1_index.py` (check 5) and `lean/check_satellite.py` (check 7): the E_1 chart step and the degree-versus-pole-order distinction; executed checks, grade B.
- `base_point_at_infinity_check.py` (check 1) and `restatement_checks.py` (check 2): the binary-form step of Lemma 2.1 and the non-Keller example of O6; executed checks, grade B.

7 Computational archive

All scripts used in this work are archived in `scripts/` with their outputs in `outputs/`, as archived with the October 1 draft:

- 18 Python scripts (SymPy-based), each with a header stating its epistemic status;
- 18 output files recording exact stdout and exit codes;
- All scripts exited with code 0 on re-execution (October 1, 2026).

This revision changes no script and claims no new re-execution: the corrections are to the statements the scripts are read as certifying (§6), not to the scripts themselves.

Reproducibility

All artifacts are archived in the public repository, directory `branch_two_dicritical/`. Each script runs with `~/miniconda3/envs/physics/bin/python` (Python 3.12, SymPy). The verification script `verify_all.sh` re-runs every script and checks exit codes.

8 Computational and AI-assisted research disclosure

Muse (Meta), the author's personal AI assistant, assisted with computational scripting, manuscript drafting, and proof verification; all script outputs were verified by actual execution. The author (B. D. McCrary) directed the investigation, verified the results, and takes responsibility for the manuscript. This revision was drafted with the same assistance: the October 1 draft was corrected against the October 13, 2026 restatement note, the derivations were expanded step by step, and no new theorem, computation, or script output was introduced. The mathematical argument stands on the artifacts above, not on these attributions.

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